A First Course in Abstract Algebra

Section 14: Factor Groups

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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14 Factor Groups

Conventions. Permutations compose from right to left. The identity is e. A group generated by an element a is denoted $\langle a \rangle$.

Exercise 14.1: Order of $\mathbb{Z}_6/\langle 3 \rangle$

The subgroup is $\langle 3 \rangle = \{0, 3\}$, of order 2. The quotient order is the number of cosets, $6/2 = 3$. Explicitly the cosets are $\{0, 3\}$, $\{1, 4\}$, $\{2, 5\}$.

Exercise 14.2: Quotient by a product subgroup

In $\mathbb{Z}_4$, $\langle 2 \rangle$ has order 2; in $\mathbb{Z}_{12}$, $\langle 2 \rangle$ has order 6. Their product therefore has order $2 \cdot 6 = 12$. The ambient product has order $4 \cdot 12 = 48$, so the quotient order is $48/12 = 4$. All groups here are abelian, so the subgroup is normal and the quotient is defined.

Exercise 14.3: Quotient by $\langle (2, 1) \rangle$

In $\mathbb{Z}_4 \times \mathbb{Z}_2$, the coordinates 2 and 1 each have order 2, so $(2, 1)$ has order 2 and generates $\{(0, 0), (2, 1)\}$. The ambient order is $4 \cdot 2 = 8$. Thus the quotient has order $8/2 = 4$.

Exercise 14.4: Quotient by the second coordinate factor

The subgroup $\{0\} \times \mathbb{Z}_5$ has five elements in the fifteen-element group $\mathbb{Z}_3 \times \mathbb{Z}_5$, so the quotient has order 3. Equivalently the projection $(a, b) \mapsto a$ has this subgroup as kernel and image $\mathbb{Z}_3$.

Exercise 14.5: Quotient of $\mathbb{Z}_2 times \mathbb{Z}_4$ by a diagonal cycle

The element $(1, 1)$ has order $\operatorname{lcm}(2, 4) = 4$, so its generated subgroup has four elements. The product has 8 elements, giving a quotient of order $8/4 = 2$.

Exercise 14.6: Quotient of $\mathbb{Z}_{12} times \mathbb{Z}_{18}$

The order of $4 \in \mathbb{Z}_{12}$ is $12/\gcd(12, 4) = 3$, and that of $3 \in \mathbb{Z}_{18}$ is $18/\gcd(18, 3) = 6$. Thus $(4, 3)$ has order $\operatorname{lcm}(3, 6) = 6$. Dividing the ambient order $12 \cdot 18 = 216$ by this subgroup order gives quotient order $216/6 = 36$.

Exercise 14.7: A quotient of $\mathbb{Z}_2 times S_3$

The element $(1, \rho_1)$ has order $\operatorname{lcm}(2, 3) = 6$. Its subgroup equals $\mathbb{Z}_2 \times A_3$: both have six elements, and all its powers lie there. This is normal because $A_3 \triangleleft S_3$ and the first coordinate is unrestricted. The ambient order is $2 \cdot 6 = 12$, so the quotient has order $12/6 = 2$. Normality matters here because the ambient group is not abelian.

Exercise 14.8: A diagonal subgroup of a coprime product

The element $(1,1)$ in $\mathbb{Z}_{11} \times \mathbb{Z}_{15}$ has order $\operatorname{lcm}(11,15) = 165$, equal to the ambient order. It therefore generates the whole group, so the quotient has just one element. Its order is 1.

Exercise 14.9: Order of $5 + \langle 4 \rangle$ in a cyclic quotient

The subgroup of $\mathbb{Z}_{12}$ is $H = \{0, 4, 8\}$. The order of $5 + H$ is the least positive k with $5k \in H$. Testing multiples modulo 12 gives $5, 10, 3, 8$ at $k = 1, 2, 3, 4$, respectively; the first three are outside H and 8 is inside. Thus the order is 4. Equivalently reduce to $5 \equiv 1$ modulo 4.

Exercise 14.10: Order of $26 + \langle 12 \rangle$

The quotient $\mathbb{Z}_{60}/\langle 12 \rangle$ is identified with $\mathbb{Z}_{12}$ by reduction modulo 12, because the kernel consists exactly of the multiples of 12 modulo 60. The coset of 26 corresponds to $2 \in \mathbb{Z}_{12}$, whose order is $12/\gcd(12,2) = 6$. Thus the requested order is 6.

Exercise 14.11: Order of a coset in $\mathbb{Z}_3 times \mathbb{Z}_6$

Let $H = \langle (1,1) \rangle$. Its elements are $(t \bmod 3, t \bmod 6)$. A multiple $k(2,1)$ lies in H if and only if an integer t satisfies

$$t \equiv 2k \pmod 3, \qquad t \equiv k \pmod 6.$$

The second congruence implies $t \equiv k \pmod 3$, so consistency is exactly $3 \mid k$. This condition is also sufficient by taking $t = k$. Therefore the least positive such k is 3, giving order 3.

Exercise 14.12: Order of a coset in $\mathbb{Z}_4^2$

The homomorphism $(x, y) \mapsto x - y \pmod 4$ has kernel $\langle (1,1) \rangle$, so the coset of $(3,1)$ corresponds to $3 - 1 = 2 \in \mathbb{Z}_4$. This element has order 2. Directly, $(3,1)$ is not on the diagonal, but $2(3,1) = (2,2)$ is, giving the same least positive return.

Exercise 14.13: Order modulo $\langle (0,2) \rangle$

In $\mathbb{Z}_4 \times \mathbb{Z}_8$, the subgroup consists of $(0,0), (0,2), (0,4), (0,6)$. For $k(3,1)$ to lie in it, the first coordinate must vanish, requiring $4 \mid 3k$ or $4 \mid k$, and the second coordinate must be even, requiring $2 \mid k$. The combined least positive solution is $k = 4$. Hence the coset has order 4.

Exercise 14.14: Order modulo $\langle (1,2) \rangle$

In $\mathbb{Z}_4 \times \mathbb{Z}_8$, use $f(x,y) = y - 2x \pmod 8$. This is well defined: changing x by 4 changes $-2x$ by -8. Its kernel is exactly $\{(t, 2t) : t \in \mathbb{Z}_4\} = \langle (1,2) \rangle$. Thus the coset of $(3,3)$ has the same order as $f(3,3) = 3 - 6 = -3 \equiv 5 \pmod 8$. Since $\gcd(5,8) = 1$, its order is 8. This also verifies that no smaller positive multiple enters the subgroup.

Exercise 14.15: A coset already equal to the identity

In $\mathbb{Z}_6 \times \mathbb{Z}_8$ one has $2(4,4) = (8,8) = (2,0)$. Hence $(2,0)$ already belongs to $H = \langle (4,4) \rangle$. The coset $(2,0) + H$ is therefore H, the identity of the quotient, and its order is 1.

Exercise 14.16: Conjugating a reflection subgroup of S_3

The source convention is $i_g(x) = gxg^{-1}$. Here $\rho_1 = (1,2,3)$ and $\mu_1 = (2,3)$. Conjugation relabels a transposition by applying the conjugating permutation to its entries:

$$\rho_1(2,3)\rho_1^{-1} = (\rho_1(2), \rho_1(3)) = (3,1) = \mu_2.$$

The identity remains the identity, so $i_{\rho_1}[H] = \{\rho_0, \mu_2\}$.

Exercise 14.17: Correcting the first normal-subgroup definition

The definition must read: a subgroup $H \leq G$ is normal if $gH = Hg$ for every $g \in G$. The original condition $hG = Gh$ for $h \in H$ says nothing about normality, because for any $h \in G$ both hG and Gh are just the whole group G. The subgroup being translated must be H, and the quantifier must range over all of G.

Exercise 14.18: The conjugation definition of normality

The stated definition is correct, provided H is a subgroup: $g^{-1}hg \in H$ for every $h \in H$ and $g \in G$ characterizes normality. Although this initially states an inclusion $g^{-1}Hg \subseteq H$, replacing g by g^{-1} gives the reverse inclusion after conjugation, so equality follows. Thus it is equivalent to invariance under every inner automorphism.

Exercise 14.19: Definition of an automorphism

An automorphism of G is an *isomorphism* from G onto itself, meaning a bijective homomorphism $G \to G$. A homomorphism merely mapping G into itself is an endomorphism and need not be invertible. For example, the constant map to e is an endomorphism of every group but is not an automorphism unless the group is trivial.

Exercise 14.20: Why normal subgroups matter

Normality is exactly the condition making the rule $(aH)(bH) = (ab)H$ independent of the chosen representatives, so that the cosets form the factor group G/H. Every normal subgroup is the kernel of the canonical homomorphism $g \mapsto gH$, and conversely every kernel is normal. Thus normal subgroups allow us to collapse specified elements to the identity while retaining a valid group operation, and describe the information lost by homomorphisms.

Exercise 14.21: Writing a proof about elements of a quotient

(a) The instructor suspects that the writer is confusing elements of G with elements of G/H, which are cosets. Naming quotient elements a, b is formally permissible, but then one cannot invoke commutativity of G on those objects without choosing representatives. The difficulty is not the letter names themselves; it is failure to track which group they belong to. **(b)** A clear beginning is: "Let $aH, bH \in G/H$, where $a, b \in G$. We must show $(aH)(bH) = (bH)(aH)$." **(c)** Since G is abelian, $ab = ba$. Therefore

$$(aH)(bH) = (ab)H = (ba)H = (bH)(aH).$$

The quotient multiplication is valid because H is normal. This proves G/H is abelian.

Exercise 14.22: A quotient of a torsion group

(a) As in Exercise 21, the likely error is treating a coset as though it were an element of the original group. The sentence "let $x \in G/H$" is not itself false, but the torsion hypothesis is about representatives in G, so the proof must display one. **(b)** Write: "Let $aH \in G/H$ with $a \in G$. Since G is torsion, a has finite order r." **(c)** Then $(aH)^r = a^r H = eH = H$, the quotient identity. Thus aH has finite order dividing r. Since the coset was arbitrary, the quotient is torsion. Its order need not equal the representative's order; it can decrease.

Exercise 14.23: True or false

(a) True: the standard coset product defines a group exactly when $N \triangleleft G$.

(b) True: $ghg^{-1} = h$ in an abelian group.

(c) True: conjugating x by g gives $gxg^{-1} = x$ when the group is abelian.

(d) True: its order is $|G|/|H|$.

(e) True: if $a^r = e$, then $(aH)^r = H$.

(f) False: $\mathbb{Z}$ is torsion free but $\mathbb{Z}/2\mathbb{Z}$ has a nonidentity element of order 2.

(g) True: $(aH)(bH) = abH = baH = (bH)(aH)$.

(h) False: $S_3/A_3 \cong \mathbb{Z}_2$ is abelian, although S_3 is not.

(i) True for positive integer n: the class of 1 generates all n residue classes.

(j) False in general: for nonzero integer n, $n\mathbb{R} = \mathbb{R}$, since $x = n(x/n)$, so the quotient is trivial, not of order n unless $n = 1$.

Exercise 14.24: The alternating subgroup and its quotient

For $n \geq 2$, the sign map $\operatorname{sgn} : S_n \to \{1, -1\}$ is a surjective homomorphism with kernel A_n. A kernel is normal, so $A_n \triangleleft S_n$, and the homomorphism theorem gives

$$S_n/A_n \cong \{1, -1\} \cong \mathbb{Z}_2.$$

Equivalently there are two cosets, the even and the odd permutations; odd times odd is even. For $n = 1$, $S_1 = A_1$ and the quotient is trivial, which is the small-case exception to the order-2 conclusion.

Exercise 14.25: Well-defined coset multiplication forces an inclusion

Let $h \in H$. Since $hH = H = eH$, well-definedness of multiplication gives

$$(ha)H = (hH)(aH) = (eH)(aH) = aH.$$

The element ha lies in its own left coset $(ha)H$, so $ha \in aH$. Since this holds for every $h \in H$, it proves $Ha \subseteq aH$. This is the desired missing direction and uses equality of cosets rather than an unproved normality assumption.

Exercise 14.26: Quotienting out all torsion

The torsion elements form a subgroup T by the earlier proof for abelian groups, and every subgroup of an abelian group is normal. Suppose gT has finite order in G/T. Then $(gT)^n = T$ for some positive n, so $g^n \in T$. Because that element is torsion, $(g^n)^m = e$ for some positive m. Thus $g^{nm} = e$, so $g \in T$ and $gT = T$. The quotient has no nonidentity torsion elements, which means it is torsion free.

Exercise 14.27: Conjugacy is an equivalence relation on subgroups

Reflexivity follows from $H = eHe^{-1}$. For symmetry, if $K = gHg^{-1}$ then $H = g^{-1}Kg$. For transitivity, if $K = gHg^{-1}$ and $L = hKh^{-1}$, then

$$L = hgHg^{-1}h^{-1} = (hg)H(hg)^{-1}.$$

Thus conjugacy of subgroups is reflexive, symmetric, and transitive. Each conjugation maps subgroups to subgroups, since it is an automorphism.

Exercise 14.28: Normal subgroups as conjugacy classes

A subgroup H is normal exactly when its conjugacy class among subgroups is the singleton $\{H\}$. Indeed, normality means $gHg^{-1} = H$ for every g, so there are no other conjugate subgroups. Conversely a singleton class means every conjugate equals H, which is the conjugation criterion for normality. This is a class of subgroups, not a conjugacy class of individual group elements.

Exercise 14.29: Conjugate reflection subgroups in S_3

The subgroup $\{\rho_0, \mu_2\}$ is generated by the transposition $(1, 3)$. Conjugating a transposition gives $(g(1), g(3))$, and as g ranges over S_3 this realizes every transposition. Hence the conjugate subgroups are exactly

$$\{\rho_0, \mu_1\}, \quad \{\rho_0, \mu_2\}, \quad \{\rho_0, \mu_3\}.$$

There are no others, since any conjugate subgroup has two elements and its nonidentity element is a transposition.

Exercise 14.30: Powers lying in a normal subgroup of finite index

Interpret $m = [G : H]$ as finite, as required for the integer exponent in the statement. The factor group has order m, so Lagrange gives $(aH)^m = H$ for each coset. Since $(aH)^m = a^m H$, this is $a^m H = H$, equivalent to $a^m \in H$. The ambient group itself need not be finite; finite index is enough. If the index is infinite, the displayed power a^m is not an ordinary group power and the statement needs the finite-index qualification.

Exercise 14.31: Intersections of normal subgroups

Let $N = \bigcap_{i \in I} N_i$ with each $N_i \triangleleft G$. It is a subgroup: all identities belong, and if $a, b \in N$, then ab^{-1} belongs to every N_i, hence to N. For $g \in G$ and $x \in N$, normality of every N_i gives $gxg^{-1} \in N_i$ for all i, so $gxg^{-1} \in N$. Thus N is normal. For an empty family the intersection is understood to be G, which is also normal.

Exercise 14.32: The normal closure of a subset

Let $\mathcal{N}$ be the collection of normal subgroups containing S. It is nonempty because G itself belongs to it. Set

$$N_S = \bigcap_{N \in \mathcal{N}} N.$$

By Exercise 31 this is normal. It contains S, since every intersected subgroup does. If M is any normal subgroup containing S, then $M \in \mathcal{N}$ and $N_S \subseteq M$ by the definition of intersection. Thus it is the unique smallest normal subgroup containing S, called the normal closure.

Exercise 14.33: The commutator quotient is abelian

By definition every commutator $aba^{-1}b^{-1}$ lies in the normal subgroup C. Therefore in G/C,

$$(aC)(bC)(aC)^{-1}(bC)^{-1} = (aba^{-1}b^{-1})C = C.$$

Multiplying by $(bC)(aC)$ on the right gives $(aC)(bC) = (bC)(aC)$. All quotient elements are such cosets, so the quotient is abelian. This proof uses only the inclusion of all commutators, not that each element of C must itself be a single commutator.

Exercise 14.34: A unique subgroup of a given order is normal

For any $g \in G$, conjugation is an automorphism, so gHg^{-1} is a subgroup of the same order as H. If H is the only subgroup of that order, $gHg^{-1} = H$ for every g. This is exactly normality. Finiteness ensures that "order" in the premise is the usual integer cardinality, although the uniqueness argument itself works for arbitrary cardinality.

Exercise 14.35: Intersecting with a normal subgroup

For $h \in H$ and $x \in H \cap N$, the conjugate hxh^{-1} lies in H because H is a subgroup, and in N because $N \triangleleft G$. Thus it lies in $H \cap N$, proving $H \cap N \triangleleft H$. It need not be normal in G: take $G = N = S_3$ and

$H = \{e, (1,2)\}$. Then N is normal in G and $H \cap N = H$. But conjugating $(1,2)$ by $(1,2,3)$ gives $(2,3) \notin H$, so this intersection is not normal in G.

Exercise 14.36: Intersection of all subgroups of a fixed finite order

Let K be the intersection of all subgroups of order s; the family is nonempty by hypothesis. Conjugation by a fixed $g \in G$ permutes that family, since $H \mapsto gHg^{-1}$ preserves subgroup order and has inverse conjugation by g^{-1}. Thus

$$gKg^{-1} = g\left(\bigcap_{|H|=s} H\right)g^{-1} = \bigcap_{|H|=s} gHg^{-1} = \bigcap_{|H|=s} H = K.$$

Conjugation commutes with intersection because it is a bijection. Since an intersection of subgroups is a subgroup, the equality proves $K \triangleleft G$ even though the individual subgroups need not be normal.

Exercise 14.37: Automorphisms and inner automorphisms

(a) The identity is an automorphism. A composite of two bijective homomorphisms is again a bijective homomorphism. The inverse of a bijective homomorphism is a homomorphism: writing $x = \phi(a)$, $y = \phi(b)$ gives $\phi^{-1}(xy) = \phi^{-1}(\phi(ab)) = ab = \phi^{-1}(x)\phi^{-1}(y)$. Composition is associative because function composition is. Thus $\mathrm{Aut}(G)$ is a group. **(b)** First prove the subgroup requirement: $i_g i_h = i_{gh}$, i_e is the identity, and $(i_g)^{-1} = i_{g^{-1}}$, so $\mathrm{Inn}(G) = \{i_g : g \in G\}$ is a subgroup. For any $\phi \in \mathrm{Aut}(G)$ and $x \in G$,

$$(\phi i_g \phi^{-1})(x) = \phi(g\phi^{-1}(x)g^{-1}) = \phi(g)x\phi(g)^{-1} = i_{\phi(g)}(x).$$

Thus conjugation by every automorphism sends inner automorphisms to inner automorphisms, proving $\mathrm{Inn}(G) \triangleleft \mathrm{Aut}(G)$.

Exercise 14.38: Elements inducing the identity inner automorphism

The condition $i_g = \mathrm{id}$ means $gxg^{-1} = x$ for every $x \in G$, equivalent to $gx = xg$ for every x. Thus the set is the center $Z(G)$. It is normal because it is the kernel of the homomorphism

$$G \to \mathrm{Aut}(G), \qquad g \mapsto i_g,$$

whose homomorphism law is $i_g i_h = i_{gh}$. Alternatively, conjugating a central element leaves it unchanged, directly giving normality.

Exercise 14.39: An induced map on factor groups

Define $\phi_* : G/H \to G'/H'$ by $\phi_*(gH) = \phi(g)H'$.

1. Check representatives.

If $gH = kH$, then $k^{-1}g \in H$. By the assumption $\phi[H] \subseteq H'$, its image $\phi(k)^{-1}\phi(g)$ is in H'. Thus $\phi(g)H' = \phi(k)H'$ and the proposed map is well defined.

2. Check products.

Normality makes both quotient products valid, and

$$\phi_*((gH)(kH)) = \phi(gk)H' = \phi(g)\phi(k)H' = (\phi(g)H')(\phi(k)H').$$

Therefore ϕ_* is a homomorphism. It is natural because it sends a coset to the coset of the original image without any arbitrary choices.

Exercise 14.40: Normal subgroups defined by determinants

(a) The determinant is a homomorphism $\det : GL(n, \mathbb{R}) \to \mathbb{R}^*$ because $\det(AB) = \det A \det B$ and invertibility gives nonzero determinant. The determinant-1 matrices form its kernel, hence a normal subgroup. Explicitly products and inverses keep determinant 1, and $\det(BAB^{-1}) = \det A$. **(b)** The map $A \mapsto |\det A|$ is a homomorphism to $\mathbb{R}^+$, because absolute value is multiplicative. Its kernel is exactly the matrices with determinant $+1$ or -1. Therefore this set is also a normal subgroup. The identity belongs to both sets because $\det I_n = 1$.

Exercise 14.41: Multiplication of subsets and the quotient group

(a) The set product is associative: $(AB)C$ and $A(BC)$ both consist of all abc with $a \in A, b \in B, c \in C$, since multiplication in G is associative. The identity subset is $\{e\}$, because $\{e\}A = A\{e\} = A$. But the powerset is not a group: the empty set has product $\varnothing$ with every subset and cannot have product $\{e\}$ with an inverse. **(b)** If $N \triangleleft G$, then for $a, b \in G$,

$$(aN)(bN) = \{an_1bn_2 : n_1, n_2 \in N\} = abN.$$

For inclusion in abN, rewrite $an_1bn_2 = ab(b^{-1}n_1b)n_2$, whose final factor lies in N. Conversely every abn is the product of $a \in aN$ and $bn \in bN$, so equality holds. Thus cosets are closed under the subset product, and the rule agrees with representative multiplication. **(c)** Associativity is inherited from subset multiplication. The subset N is an identity on cosets, since $N(aN) = aN = (aN)N$ by the displayed equality and normality. Each aN has inverse $a^{-1}N$, because their products in both orders equal N. Hence the cosets form a group, proved directly from subsets. Its identity is N, whereas the identity of the whole powerset monoid is $\{e\}$; they coincide exactly when $N = \{e\}$.